

Physics for mathematicians: course sample

PROBLEM • physics:foundations:balances:pb:point-flux

Flux detects a point source

For $x \neq 0$, let $j(x) = qx/(4\pi|x|^3)$. Show that $\nabla \cdot j = 0$ away from zero and that its distributional divergence on $\mathbb{R}^3$ is $q\delta_0$.

Solution

For $r = |x| > 0$, $\nabla \cdot (xr^{-3}) = 3r^{-3} - 3r^{-5}|x|^2 = 0$. On a sphere of radius ϵ , $j \cdot n = q/(4\pi\epsilon^2)$, so the outward flux is q .

For a test function f , integrate $\nabla \cdot (fj) = j \cdot \nabla f$ on the complement of the radius- ϵ ball, truncating outside the support of f . Its inner boundary normal points inward, so $-\int_{|x|>\epsilon} j \cdot \nabla f = q(4\pi\epsilon^2)^{-1} \int_{|x|=\epsilon} f dA$. The right side tends to $qf(0)$, and the locally integrable field permits the left limit. This is the distributional assertion.

Response of a stable linear system

Let A be a real matrix with all eigenvalues of negative real part and let b be a fixed vector. For $\dot{x} = Ax + bu(t)$ with scalar continuous input, the solution is

$$x(t) = e^{A(t-t_0)}x(t_0) + \int_{t_0}^t e^{A(t-s)}bu(s) ds.$$

For bounded continuous input on the whole real line, the unique solution bounded for all time is $x(t) = \int_{-\infty}^t e^{A(t-s)}bu(s) ds$.

Proof

Multiplying the equation by e^{-At} and integrating gives the first formula; differentiation verifies it. Jordan normal form gives $\|e^{Ar}\| \leq Ce^{-\alpha r}$ for $r \geq 0$ and some $C, \alpha > 0$, so the improper integral converges uniformly under bounded input and is bounded. The difference of two bounded solutions is $e^{A(t-t_0)}v$. If it were nonzero, boundedness in the remote past would fail: $v = e^{AR}x(t_0 - R)$ tends to zero as $R \rightarrow \infty$ for bounded x . Thus $v = 0$.

Observable evolution and a conserved bracket

Prerequisites: @physics:classical:hamilton:th:symplectic-volume.

For smooth $f(q, p, t)$ along a Hamiltonian trajectory,

$$\frac{df}{dt} = \partial_t f + \{f, H\}.$$

If two time-independent observables f, g satisfy $\{f, H\} = \{g, H\} = 0$ for a smooth autonomous Hamiltonian, then $\{f, g\}$ is also conserved.

Proof

The first identity follows by substituting Hamilton's equations into the chain rule. To prove the second without appealing to an unstated bracket identity, let Φ_t be the Hamilton flow. For a covector β , define X_β by $i_{X_\beta}\omega = \beta$. Preservation of ω preserves the induced pairing $(\alpha, \beta) \mapsto \alpha(X_\beta)$, which is exactly the displayed canonical bracket: $\{f, g\} \circ \Phi_t = \{f \circ \Phi_t, g \circ \Phi_t\}$. This follows by inserting the derivatives of the pullbacks in that pairing and using $\Phi_t^*\omega = \omega$. The first identity gives $f \circ \Phi_t = f$ and $g \circ \Phi_t = g$. Therefore $\{f, g\} \circ \Phi_t = \{f, g\}$, as required.

First frequency correction for a cubic restoring force

Consider $\ddot{x} + \Omega_0^2 x + \alpha x^3 = 0$, with $\Omega_0, A > 0$, $x(0) = A$, $\dot{x}(0) = 0$. Set $\epsilon = \alpha A^2 / \Omega_0^2$ and assume $|\epsilon| \ll 1$. Seek $x = A[\cos \tau + \epsilon y_1(\tau)]$, $\tau = \Omega_0(1 + \epsilon \nu)t$, with $y_1(0) = y_1'(0) = 0$. Remove resonant forcing at first order, and state exactly what has been checked.

Solution

Divide by $A\Omega_0^2$ and expand to first order. The equation for y_1 is $y_1'' + y_1 = 2\nu \cos \tau - \cos^3 \tau = (2\nu - 3/4) \cos \tau - (1/4) \cos 3\tau$. Resonance is absent when $\nu = 3/8$. Then $y_1 = (\cos 3\tau - \cos \tau)/32$ satisfies both the equation and the initial data. Thus the formal frequency is $\Omega_0[1 + 3\epsilon/8]$.

Substitution into the original differential operator leaves a bounded residual of order $A\Omega_0^2\epsilon^2$ for small ϵ , since the omitted terms are products of bounded trigonometric functions. The initial displacement and velocity are exact for this approximation. A small residual is not a proof of a uniformly small trajectory error over arbitrarily long times. Positive α raises this first frequency correction; $[\alpha] = \text{m}^{-2} \text{s}^{-2}$.

A plucked string and its harmonic content

Prerequisites: @physics:waves:mechanical-waves:th:fixed-string-modes.

A string of length L is initially at rest with triangular displacement $f(x) = 2Ax/L$ for $0 \leq x \leq L/2$ and $f(x) = 2A(L-x)/L$ for $L/2 \leq x \leq L$. Assume $2|A|/L \ll 1$. Find its sine coefficients, write the mode evolution, and identify why a globally C^2 solution is not obtained from these initial data.

Solution

The coefficients are $b_n = (2/L) \int_0^L f(x) \sin(n\pi x/L) dx$. Integration by parts on each half cancels endpoint terms and yields $b_n = 8A \sin(n\pi/2)/(n^2\pi^2)$. Thus the modal solution is $y(x, t) = \sum_{n \geq 1} b_n \sin(n\pi x/L) \cos(n\pi ct/L)$. Since $\sum |b_n| < \infty$, this series converges uniformly in (x, t) and reproduces the continuous initial displacement by its Fourier series. Its modes have zero initial velocity; in the finite-energy interpretation the velocity is zero initially as well, since the squared velocity coefficients are dominated by a summable multiple of $1/n^2$ and vanish termwise at $t = 0$. Every partial sum solves the wave equation; uniform convergence allows that equation to pass to the limit in distributions by moving derivatives onto a compactly supported smooth test function. Also $\sum n^2 b_n^2 < \infty$, so the spatial derivative has finite squared integral by sine/cosine orthogonality. The initial slope jumps at the midpoint, so initial displacement is not even C^1 there. The series describes a weak finite-energy wave with traveling corners, not a globally C^2 field.

Snell refraction from tangential phase matching

Prerequisites: `@physics:waves:interfaces:ax:interface-models`.

Under the stationary plane optical-interface model, a reflected traveling wave has equal incidence and reflection angles, and a transmitted traveling wave satisfies

$$n_1 \sin \theta_1 = n_2 \sin \theta_2.$$

Angles are measured from the interface normal in the incidence plane. This equation is conditional on a real transmitted traveling wave existing.

Proof

Boundary equality for all times requires equal temporal frequency, since exponentials with distinct frequencies cannot have the same nonzero single-frequency trace. Equality along every point of the plane similarly fixes the tangential component of wave vector. Reflection remains in medium 1 with the same magnitude of $\mathbf{k}$, so its tangential component fixes the same angle magnitude; the normal component reverses sign. Transmission has tangential magnitudes $(n_j \omega / c_0) \sin \theta_j$. Their equality gives Snell's relation after cancellation. If the required sine exceeds one, the traveling-wave assumption fails rather than giving a real angle.

A charge above a grounded conducting plane

A charge q sits at $(0, 0, a)$ with $a > 0$, above a grounded infinite conductor occupying $z \leq 0$. Find a potential for $z > 0$ that vanishes on the plane and at infinity. Determine the induced surface charge density and its total charge. Verify uniqueness among potentials with the same Coulomb singularity and uniform decay at infinity.

Solution

Place an auxiliary charge $-q$ at $(0, 0, -a)$ outside the physical domain. With $R_{\pm} = \sqrt{x^2 + y^2 + (z \mp a)^2}$, take $\phi = q(R_+^{-1} - R_-^{-1})/(4\pi\epsilon_0)$. It has the correct singularity, is harmonic elsewhere in $z > 0$, and is zero when $z = 0$.

Using the outward normal from the conductor, $\sigma = \epsilon_0 E_z(0^+) = -qa/[2\pi(s^2 + a^2)^{3/2}]$, where $s = \sqrt{x^2 + y^2}$. Thus

$$Q_{\text{induced}} = 2\pi \int_0^\infty \sigma(s) s \, ds = -qa \int_0^\infty \frac{s \, ds}{(s^2 + a^2)^{3/2}} = -q.$$

For uniqueness, the difference of two candidates extends harmonically across the charge since their singular parts cancel. On a large half-ball it is zero on the flat boundary and bounded in absolute value on the hemisphere by a number tending to zero. The maximum principle applied to the difference and its negative gives zero at every fixed interior point. The auxiliary charge represents boundary data; it is not a second physical charge above the conductor.

Interface jump conditions with surface sources

Across a fixed smooth interface with normal $\mathbf{n}$ from medium 1 to 2, assume piecewise smooth finite fields, free surface charge σ_f , free surface current $\mathbf{K}_f$, and no delta-function magnetic flux or electric displacement sheets. Maxwell's equations imply

$$\mathbf{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \sigma_f, \quad \mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0,$$

$$\mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) = 0, \quad \mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{K}_f.$$

Proof

Integrate the two divergence equations over a thin pillbox crossing the interface. As its thickness tends to zero, the side flux vanishes, and its top-minus-bottom flux per area is the normal jump. Free volume charge contributes only its surface-sheet integral in this limit, giving the first two equations.

Integrate Faraday and Ampère–Maxwell around a thin rectangle crossing the interface. The two short sides vanish in the limit. Finite time derivatives of the fields contribute vanishing flux through its shrinking area. Faraday therefore gives zero tangential electric jump; the free sheet current contributes a finite term to Ampère's law. With the right-hand orientation, that term is exactly $\mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{K}_f$. Doing this for every tangential rectangle direction gives the vector equations.

Entropy stationarity under energy exchange

Two subsystems have differentiable entropies $S_1(U_1)$, $S_2(U_2)$, fixed volumes and particle counts, and total energy U . If an interior entropy maximum occurs under energy exchange and both temperatures are positive, then $T_1 = T_2$. If the entropies are twice differentiable and both heat capacities are positive, any such stationary point is a strict local maximum.

Proof

Write $s(u) = S_1(u) + S_2(U - u)$. The fundamental relation implies $s'(u) = 1/T_1 - 1/T_2$, so stationarity gives equal temperatures. Differentiating $S'_i = 1/T_i$ yields $S''_i = -1/(T_i^2 C_{V,i})$. At the stationary point, $s'' = S''_1 + S''_2 < 0$ if both capacities are positive. The second derivative test proves strict local maximality. Endpoint maxima need not satisfy the interior derivative condition.

Free expansion versus reversible expansion

A calorically perfect ideal gas expands into a vacuum inside an insulated rigid enclosure, from equilibrium volume V_1 to $V_2 > V_1$. Find $Q, W, \Delta U, \Delta T, \Delta S$. Compare with reversible isothermal expansion between the same endpoint states.

Solution

The enclosure receives no heat and the gas pushes against zero external pressure: $Q = W = 0$. The first law gives $\Delta U = 0$, so $cNk_B\Delta T = 0$ and the final temperature is unchanged. The equilibrium entropy difference is $Nk_B \log(V_2/V_1) > 0$. A reversible isothermal expansion instead produces $W = \int_{V_1}^{V_2} Nk_B T dV/V = Nk_B T \log(V_2/V_1)$ and needs $Q = W$ to maintain constant U . The entropy change is the same state difference, but its origin is reversible heat input in the second process and entropy production in the first.

A quantum mode and the classical limit

A harmonic mode has energies $E_j = \hbar\omega(j + 1/2)$ for $j \geq 0$, with $\omega > 0$ fixed. Find its partition function, mean energy, and heat capacity, including the two temperature limits.

Solution

Let $x = \beta\hbar\omega > 0$. Geometric summation gives $Z = e^{-x/2}/(1 - e^{-x})$. Differentiating its logarithm gives $U = \hbar\omega/2 + \hbar\omega/(e^x - 1)$. Differentiating U yields

$$C_V = k_B x^2 \frac{e^x}{(e^x - 1)^2}.$$

As $T \downarrow 0$, $U \rightarrow \hbar\omega/2$ and $C_V \sim k_B x^2 e^{-x}$. As $T \rightarrow \infty$, $U \sim k_B T$ and $C_V \rightarrow k_B$, recovering two classical quadratic terms. The zero-point energy is temperature-independent and contributes no heat capacity. The partition function satisfies $Z \sim e^{-x/2}$ as $T \downarrow 0$, while $Z \sim 1/x = k_B T/(\hbar\omega)$ as $T \rightarrow \infty$.

Detailed balance decreases relative entropy

For a finite detailed-balance Markov model with $\pi_i > 0$ and $\sum_i \pi_i = 1$, along a differentiable solution with every $p_i > 0$,

$$D(p\|\pi) = \sum_i p_i \log(p_i/\pi_i)$$

is nonincreasing. If the undirected positive-rate graph is connected, equality of its time derivative occurs only at $p = \pi$.

Proof

Put $r_i = p_i/\pi_i$ and $a_{ij} = w_{ij}\pi_j = a_{ji}$. Conservation of probability cancels the derivative of the linear terms, so

$$\dot{D} = \sum_{i,j \neq i} a_{ij}(r_j - r_i) \log r_i = -\frac{1}{2} \sum_{i,j \neq i} a_{ij}(r_i - r_j)(\log r_i - \log r_j) \leq 0.$$

The logarithm is strictly increasing. Each nonzero-rate summand vanishes precisely when $r_i = r_j$. Connectivity therefore forces a common ratio, and normalization makes it one. This establishes monotonicity for the stated stochastic model, without deriving the model from microscopic mechanics.

The inertial clock maximizes elapsed time between fixed events

Between two timelike-separated events in Minkowski spacetime, the straight future-directed timelike segment has at least as much proper time as any future-directed piecewise smooth timelike curve joining them. Equality holds exactly for that straight image with future-directed parametrization.

Proof

Choose the inertial rest frame of the endpoint separation; the endpoints then have the same spatial position and time difference $T > 0$. Every competing future timelike curve can be parametrized by this coordinate time, since its time coordinate increases strictly. Its elapsed time is $\int_0^T \sqrt{1 - |\mathbf{v}|^2/c^2} dt \leq T$. The straight stationary segment attains T . Equality requires $\mathbf{v} = 0$ almost everywhere. Piecewise smoothness then makes each smooth piece spatially constant and continuity joins those constants, giving the same straight segment. Conversely that segment attains equality. This is a global flat-spacetime statement; curved-spacetime geodesics need not globally maximize proper time.

A round trip with two inertial legs

One ideal clock stays at $x = 0$. Another leaves it at $t = 0$, travels at speed v to $x = L > 0$, reverses direction, and returns at the same speed, where $0 < v < c$. In the idealized instantaneous-turnaround limit, compute both elapsed times and their difference. Identify what breaks the apparent exchange symmetry.

Solution

The stationary clock measures $T = 2L/v$. Each traveling leg takes laboratory time L/v and proper time $L/(v\gamma)$, so $\tau = 2L/(v\gamma)$ and $T - \tau = (2L/v)(1 - \sqrt{1 - v^2/c^2}) > 0$. These are lengths of two specified worldlines between the same endpoints, not contradictory applications of time dilation to one inertial pair. The traveler changes inertial frames at turnaround; the stationary clock does not. A smooth short turnaround approaches this piecewise timelike calculation when its additional durations are negligible. There is no universal rule that aging difference is a function only of peak acceleration.

An exact check on perturbation theory

For $\Delta > 0$ and real energy scale g , diagonalize $H = \begin{pmatrix} 0 & g \\ g & \Delta \end{pmatrix}$. Expand the lower eigenvalue for $|g| \ll \Delta$ through order g^2 and explain why that expansion fails near $\Delta = 0$.

Solution

The characteristic polynomial is $E^2 - \Delta E - g^2$, so

$$E_{\pm} = \frac{\Delta \pm \sqrt{\Delta^2 + 4g^2}}{2}.$$

For $g \neq 0$, corresponding normalized eigenvectors are

$$v_{\pm} = \frac{(g, E_{\pm})^T}{\sqrt{g^2 + E_{\pm}^2}}.$$

Multiplication by H verifies the eigenvalue equations using $E_{\pm}^2 = \Delta E_{\pm} + g^2$. Their inner product vanishes because $E_+ E_- = -g^2$. Thus the orthogonal matrix with columns v_- , v_+ diagonalizes H to $\text{diag}(E_-, E_+)$. For $g = 0$, use the standard basis, with $E_- = 0$ and $E_+ = \Delta$.

For $\Delta > 0$, the Taylor expansion $\sqrt{1+x} = 1 + x/2 + O(x^2)$ with $x = 4g^2/\Delta^2$ gives $E_- = -g^2/\Delta + O(g^4/\Delta^3)$. The correction lowers the unperturbed ground energy. This expansion assumes $|g|/\Delta \ll 1$; as the gap closes that ratio ceases to be small. At $\Delta = 0$ the exact eigenvalues are $\pm|g|$, so the nondegenerate expansion is inappropriate.

A controlled first Born coefficient for a compact potential

Let real $V \in L^\infty(\mathbb{R}^3)$ be supported in a ball $B \subset \mathbb{R}^3$, and fix $k > 0$. Consider the outgoing integral-equation model

$$\psi_\lambda(x) = e^{ikz} - \lambda C \int_B \frac{e^{ik|x-y|}}{|x-y|} V(y) \psi_\lambda(y) dy, \quad C = \frac{m}{2\pi\hbar^2}.$$

For sufficiently small real λ , it has a unique bounded solution on B , extended by this equation outside B , whose amplitude satisfies uniformly in unit $\mathbf{n}$

$$f_\lambda(\mathbf{n}) = -\lambda C \int_B e^{-ik\mathbf{n}\cdot\mathbf{y}} V(y) e^{iky_z} dy + O(\lambda^2).$$

Proof

Define $Kf(x) = C \int_B e^{ik|x-y|} |x-y|^{-1} V(y) f(y) dy$ for $x \in B$. The supremum over $x \in B$ of $\int_B |x-y|^{-1} dy$ is finite: if the ball has radius R , it is bounded by $\int_{|u| \leq 2R} |u|^{-1} du = 8\pi R^2$. Thus K is bounded on $L^\infty(B)$. If $|\lambda| \|K\| < 1$, the geometric operator series gives $(I + \lambda K)^{-1}$, proving existence, uniqueness, and $\psi_\lambda = e^{ikz} + O(\lambda)$ in that norm. For $x = r\mathbf{n}$ and bounded y , $|x-y| = r - \mathbf{n} \cdot y + O(r^{-1})$ and $|x-y|^{-1} = r^{-1} + O(r^{-2})$, uniformly in angle. Substitution into the outside integral gives $f_\lambda = -\lambda C \int_B e^{-ik\mathbf{n}\cdot\mathbf{y}} V(y) \psi_\lambda(y) dy$. Replacing ψ_λ by its norm expansion yields the asserted uniform $O(\lambda^2)$ error. This theorem treats the stated integral model and does not prove general scattering completeness.

Angular-momentum balance enforces symmetric stress

Prerequisites: @physics:continuum:balances:th:local-momentum.

Assume the preceding smooth mass and linear-momentum balances, and also angular-momentum balance with only the torques $\mathbf{x} \times \rho \mathbf{b}$ and $\mathbf{x} \times (\boldsymbol{\sigma} \mathbf{n})$, about a fixed inertial origin. Then $\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$. Conversely symmetric stress makes that angular balance follow from linear momentum. This is conditional on the absence of body couples, couple stresses, and independent spin.

Proof

The derivative of $\int_{B_t} \rho(\mathbf{x} \times \mathbf{v}) d\mathbf{x}$ is $\int_{B_t} \rho \mathbf{x} \times D_t \mathbf{v} d\mathbf{x}$, because $\mathbf{v} \times \mathbf{v} = \mathbf{0}$. Insert local linear momentum. For component k , the divergence theorem applied to $\epsilon_{k\ell m} x_\ell \sigma_{mj}$ gives the surface torque as the volume integral of $[\mathbf{x} \times \operatorname{div} \boldsymbol{\sigma}]_k + \epsilon_{kjm} \sigma_{mj}$. Thus angular balance is equivalent, for every region, to $\epsilon_{kjm} \sigma_{mj} = 0$ for $k = 1, 2, 3$. These three equations are the vanishing of the three independent antisymmetric stress components. They are equivalent to symmetry, and the calculation reverses for the converse.

Young modulus and Poisson ratio from Lamé moduli

Assume $\mu > 0$ and $3\lambda + 2\mu > 0$. Under uniform uniaxial stress $\sigma_{11} = s$, with every other stress component zero, solve for the diagonal strains and find Young's modulus $Y = s/\epsilon_{11}$ and Poisson ratio $\nu_P = -\epsilon_{22}/\epsilon_{11}$ for $s \neq 0$.

Solution

Taking the trace of Hooke's law gives $\operatorname{tr} \boldsymbol{\epsilon} = s/(3\lambda + 2\mu)$. Therefore $\epsilon_{11} = [s - \lambda s/(3\lambda + 2\mu)]/(2\mu) = s(\lambda + \mu)/[\mu(3\lambda + 2\mu)]$ and $\epsilon_{22} = \epsilon_{33} = -\lambda s/[2\mu(3\lambda + 2\mu)]$. Hence $Y = \mu(3\lambda + 2\mu)/(\lambda + \mu)$ and $\nu_P = \lambda/[2(\lambda + \mu)]$. Positivity implies $\lambda + \mu > 0$, so these expressions are well-defined, with $Y > 0$ and $-1 < \nu_P < 1/2$. Off-diagonal strains are zero because the corresponding stresses are $2\mu \epsilon_{ij}$.

Interior variation of the gravitational action

For smooth compactly supported inverse-metric variations, with matter variation defined in @physics:gravitation:einstein-matter:ax:einstein-hilbert-action, stationarity of $S_g + S_m$ at fixed matter fields is equivalent to Einstein's equation in the interior.

Proof

Write $q^{\mu\nu} = \delta g^{\mu\nu}$. Differentiate $g_{\mu\rho}g^{\rho\nu} = \delta_\mu^\nu$ to get $\delta g_{\mu\nu} = -g_{\mu\alpha}g_{\nu\beta}q^{\alpha\beta}$. The determinant derivative gives $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}q^{\mu\nu}$, where $\sqrt{-g} = \sqrt{-\det g}$.

Let $C_{\mu\nu}^\rho = \delta\Gamma_{\mu\nu}^\rho$. It is a tensor, as a variation of a connection. Differentiation of the curvature formula gives

$$\delta R_{\mu\nu} = \nabla_\rho C_{\nu\mu}^\rho - \nabla_\nu C_{\rho\mu}^\rho.$$

Indeed the partial-derivative terms are $\partial_\rho C_{\nu\mu}^\rho - \partial_\nu C_{\rho\mu}^\rho$; the four product-rule terms from the two connection products are exactly the connection corrections to these two covariant derivatives, using symmetry of their lower indices. Metric compatibility then makes $g^{\mu\nu}\delta R_{\mu\nu} = \nabla_\rho V^\rho$ for $V^\rho = g^{\mu\nu}C_{\mu\nu}^\rho - g^{\mu\rho}C_{\lambda\mu}^\lambda$. Its integral vanishes because $\sqrt{-g}\nabla_\rho V^\rho = \partial_\rho(\sqrt{-g}V^\rho)$ and V has compact support.

Vary $R = g^{\mu\nu}R_{\mu\nu}$ and the volume element. The surviving terms give

$$\delta(S_g + S_m) = \int \sqrt{-g} \left[\frac{c^3}{16\pi G} (G_{\mu\nu} + \Lambda g_{\mu\nu}) - \frac{1}{2c} T_{\mu\nu} \right] q^{\mu\nu} d^4x.$$

The coefficient is symmetric. Arbitrary compactly supported symmetric q and the fundamental lemma of variations force it to vanish pointwise. Multiplying by $16\pi G/c^3$ yields Einstein's equation. Conversely that equation makes every such first variation vanish.

Derive the leading periapsis advance

Use the Schwarzschild orbit equation $u'' + u = GM/\ell^2 + 3GMu^2/c^2$, where $u = 1/r$ and primes mean $d/d\phi$. Set $p = \ell^2/(GM)$, $GM/(pc^2) \ll 1$, and $u_0 = (1 + e_{\text{orb}} \cos \phi)/p$ with $0 < e_{\text{orb}} < 1$. Find the resonant first-order correction and leading periapsis advance. At Newtonian order $p = a(1 - e_{\text{orb}}^2)$.

Solution

Write $m_g = GM/c^2$, a length. To first order, $u_1'' + u_1 = 3m_g u_0^2$. Expanding the square gives a constant, a double-frequency term, and resonant term $(6m_g e_{\text{orb}}/p^2) \cos \phi$. Since $(d^2/d\phi^2 + 1)(\phi \sin \phi) = 2 \cos \phi$, the secular term is $u_{1,\text{sec}} = 3m_g e_{\text{orb}} \phi \sin \phi / p^2$. Constant and double-frequency terms have bounded particular solutions and do not create the leading secular phase.

Compare with $(e_{\text{orb}}/p) \cos((1 - \delta)\phi) = (e_{\text{orb}}/p)(\cos \phi + \delta \phi \sin \phi) + O(\delta^2 \phi^2)$. Thus $\delta = 3m_g/p$. Periapses have angular separation $2\pi/(1 - \delta)$, giving advance $2\pi\delta + O(\delta^2) = 6\pi GM/[a(1 - e_{\text{orb}}^2)c^2]$ to leading order. This phase interpretation is perturbative over orbital scales; the truncated secular expression is not uniformly valid for arbitrarily many orbits.

The static screened Green function

For $m > 0$ in three spatial dimensions, verify that $G(x) = e^{-m|x|}/(4\pi|x|)$ satisfies $(-\Delta + m^2)G = \delta_0$ distributionally and decays at infinity. Interpret m^{-1} .

Solution

For $r > 0$, the radial identity $\Delta(u(r)/r) = u''(r)/r$ gives $\Delta G = m^2 G$. Near zero, $G = (4\pi r)^{-1} + O(1)$ and $\partial_r G = -1/(4\pi r^2) + O(1)$. Integration by parts outside a radius- ϵ sphere against a test function therefore leaves a boundary term tending to its value at zero; the term involving G times the test-function normal derivative tends to zero. Thus $-\Delta G + m^2 G$ has unit delta strength. The exponential ensures decay; m^{-1} is the screening length in the declared natural units. A force law additionally needs the coupling and source normalization.

The Gaussian fourth-moment pairing

Let $\mathbf{X}$ be a centered real Gaussian vector of finite dimension with covariance $\mathbf{C}$. Then

$$\mathbb{E}[X_i X_j X_k X_l] = C_{ij}C_{kl} + C_{ik}C_{jl} + C_{il}C_{jk}.$$

This exact finite-dimensional identity is a simple model of pair contractions in free-field calculations.

Proof

The moment generating function is $M(\mathbf{t}) = \exp(\mathbf{t}^T \mathbf{C} \mathbf{t} / 2)$, including for singular $\mathbf{C}$ by representing $\mathbf{X}$ as a linear image of a standard Gaussian. Four derivatives at $\mathbf{t} = \mathbf{0}$ select only the quadratic term in the exponential series of the quadratic form, namely $(\mathbf{t}^T \mathbf{C} \mathbf{t})^2 / 8$. Assigning the four derivative labels to the two paired factors gives exactly the three pairings shown. Gaussian moments are finite, so differentiation under expectation is justified near zero.